

Mori-Zwanzig Mode Decomposition: Transient Flows and Laminar-Turbulent Boundary Layer Transition

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In this work, we apply and analyze the data-driven Mori-Zwanzig Mode Decomposition (MZMD), a novel technique used for extracting large-scale spatio-temporal coherent structures representing an efficient generalization of Dynamic Mode Decomposition (DMD). We first demonstrate that by adding memory terms, MZMD is able to reconstruct a 2D transient flow over a cylinder; where the flow transitions from an unstable equilibrium point to a heteroclinic orbit representing the von Kármán vortex street. For this flow regime, DMD is not able to reconstruct the flow, which we demonstrate. We next show that MZMD significantly improves upon DMD for reconstructing and performing future state predictions of a high-dimensional flow with strong nonlinearities, namely the laminar-turbulent transition in a hypersonic boundary layer of a tangent ogive with a blunted nose tip at Mach 5. For this flow, we demonstrate the diagnostic ability of MZMD for predicting and extracting the dominant mechanisms relevant to the transition to turbulence. The results improve as more memory terms are added, however the improvement saturates after a certain number of memory terms are considered, suggesting a finite memory effect. The transition to turbulence is further analyzed using the MZMD modal structures and spectrum.

I. Introduction

Modal decomposition techniques of an experimental or numerical data-set of the flow field [1] have shown promise to improve our understanding of flow physics. For example, extracting the most energetic spatial modes with proper orthogonal decomposition (POD) [2], or the spatio-temporal coherent structures with Dynamic Mode Decomposition (DMD) [3] and Spectral POD [4]. For example, transitional and turbulent flows arise in many applications, and often contain a wide range of coherent structures; the analysis of which are key in understanding the dominant physical mechanisms present in the flow. Additionally, many flows in nature contain transient dynamics, where the fluid system moves from one equilibrium state to another. Flows of each type represent significant challenges for current modal decomposition techniques. In this work we investigate the ability of MZMD to perform modal analysis and reduced order modelling of two such flows: a 3D hypersonic boundary layer transitional flow, and a 2D transient flow over a cylinder.

Dynamic mode decomposition (DMD) is a data-driven technique that has achieved wide ranging success [3]. DMD extracts large-scale spatio-temporal structures from data by computing an eigendecomposition of the linear operator that best fits the discrete time evolution of state variables. Additionally, DMD has theoretical connections to the Koopman operator [5]; however, in practice, finding a Koopman invariant subspace can be intractable. Therefore, a key issue with approximate Koopman learning techniques is that, in general, finding a finite set of observables that completely closes the dynamics is challenging. DMD also has limitations in extracting or capturing the physical mechanisms associated with intermediate transients and intermittent dynamics, which are characteristics of nonlinear systems with complex dynamics [6, 7]. Furthermore, in applications containing nonlinear dynamics, DMD may yield completely spurious results [8]. The key challenge of DMD, then, is to identify a finite-dimensional invariant subspace of the Koopman operator, ensuring that the dynamics evolving according to the Koopman operator are closed within this space.

Variants of DMD have emerged to address these issues by increasing the dimension of the functional basis that these linear models can represent by incorporating additional observables. For example, Extended DMD (EDMD) [9] includes higher-order observables, such as polynomial functions of the states alongside the states themselves. However, this approach can become extremely computationally expensive for high-dimensional dynamical systems, requires

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a priori knowledge of the system, and may lead to overfitting. Other methods have attempted to combine nonlinear auto-encoders with linear least squares to simultaneously approximate observables and the Koopman operator [10]. Nevertheless, in high-dimensional nonlinear dynamical systems, discovering a finite-dimensional invariant subspace can be exceedingly challenging [11].

Other variants have emerged that incorporate past history into the observable selection, aiming to close the dynamics in the linear Koopman space more effectively by increasing the expressive power of the functional basis represented by these linear models. Two common approaches to achieve this are time-delay embeddings [12] and Higher-Order DMD (HODMD) [8], each showing promise in addressing the aforementioned weaknesses of DMD [7]. It is worth noting that HODMD also incorporates time delay embedding, but does so within the space of observables that are the selected for by using SVD as a linear auto-encoder. Thus both methods are taking advantage of the Takens' delay embedding theorem [13]. Satisfying this theorem requires that the nonlinear manifold of dimension m , describing the state space evolution, can be embedded in a linear manifold ($\mathbb{R}^N$) provided the embedding dimension $N \geq 2m + 1$, where m is the dimension of the nonlinear manifold of the strange attractor (typically unknown a-priori). Although this embedding dimension can be less for certain cases [14], this could result in very large concatenation of the snapshot matrices involved with the Hankel DMD and HODMD formulation, thus leading to intractable computational costs for high-dimensional nonlinear dynamical systems. While these methods do not necessitate a priori selection of higher-order observables as in EDMD, they encounter similar issues, such as significantly increased computational costs for high-dimensional dynamical systems and potential overfitting.

The Mori-Zwanzig (MZ) formalism, originally developed in statistical mechanics [15, 16], offers an alternative (and complementary) framework to the Koopman theory for constructing reduced-order models. The MZ formalism utilizes a projection operator to decompose any function into a function of the resolved variables (observables) and its orthogonal complement (which depends on both the resolved and unresolved variables). The MZ procedure then derives the Generalized Langevin Equation (GLE) from the associated Liouville equation acting on the predefined observables. The GLE consists of a Markovian term, a memory term, and an orthogonal dynamics term. The GLE is a closed and exact system of evolutionary equations for the observables where the effects due to the unresolved dynamics are captured in the memory kernel and orthogonal dynamics. Therefore, the MZ procedure provides an alternative (and complementary) framework to the Koopman theory for constructing closed reduced-order models without relying on Takens' embedding theorem. Recent developments by Y.T. Lin et al. [17, 18] have led to general data-driven techniques to approximate the MZ operators from data (and seeing applications in turbulent flows [19–21]). This framework offers an attractive formulation of reduced order models that can address the aforementioned issues of approximate Koopman techniques such as EDMD, Hankle DMD and HODMD. Because the MZ framework does not rely on the Whitney embedding theorem, the MZ operators can be learned more efficiently, as shown in Woodward et al. [22].

In our recent work [22], we constructed MZMD, a novel MZ modal decomposition technique which efficiently generalizes DMD by including MZ memory terms and significantly outperforms DMD for a similar computational cost. MZMD generalized DMD in the following ways; using the same state measurements, MZMD closes the dynamics with MZ memory. We show that MZMD improves the modeling of complex flows where DMD struggles, intermediate transients and intermittent dynamics. Furthermore, MZMD includes memory without the more costly time-delay embeddings [12, 23] or Higher-Order DMD [8] (as shown in [22]). A key result of this investigation was that MZ memory and Takens' time-delay embedding represent two unique approaches for adding hysteresis information. These two approaches of adding memory are not mutually exclusive, and empirical evidence shows that mixing both time-delay embedding with MZ memory can obtain the highest accuracy with the least computational cost. This is based on the data-driven Mori-Zwanzig (MZ) approach first derived in Lin et al. [18] and further extended in [17], which generalizes the approximate Koopmanian learning and has shown to improve the prediction performance over Extended DMD (EDMD) [24]. Thus, by including MZ memory, we can still use the state space variables (which is advantageous for modal analysis) while systematically improving over the truncation errors and limitations of DMD.

In this work, we present the formulation and analysis of the MZMD modes and spectrum and compare it to DMD when applied to several Direct Numerical Simulation (DNS) data-sets where DMD struggles. We show that MZMD not only improves the ability to predict and reconstruct strongly nonlinear regions of the flow (which contain intermittent dynamics), but we also demonstrate the ability of MZMD to capture transient dynamics. Thus by adding MZ memory terms, MZMD extracts modes that are dynamically short lived, but crucial for capturing the transition from one equilibrium state to another (and intermittent dynamics that improves the resolution in nonlinear regions of the flow). To demonstrate this, the first data-set we investigate is a 2D transient flow over a cylinder at Reynolds number $Re = 200$. This data-set is obtained by using the immersed boundary method developed by Colonius and Taira [25]. We next investigate the ability of MZMD for predicting and understanding the dominant mechanisms relevant for

the laminar-turbulent transition in a hypersonic boundary layer of a tangent ogive with a blunted nose tip at Mach 5 [26]. We show that by including MZ memory, this introduces nontrivial short lived transient modes that improves the resolution of nonlinearities (containing intermittent dynamics) as well as global transient dynamics. For each dataset we investigate the importance of the MZ memory terms in MZMD as compared to DMD, and analyze their contribution to understanding the dominant mechanisms of the flow.

II. Data-Driven Mori-Zwanzig

Following Lin et al. [17, 18], we consider the discrete-time autonomous deterministic dynamical system where the states $\mathbf{x}(t) \in \mathbb{R}^D$ evolve according to

$$\mathbf{x}_{n+1} = \mathbf{F}(\mathbf{x}_n), \quad \mathbf{x}(0) = \mathbf{x}_0, \quad (1)$$

where $\mathbf{F} : \mathbb{R}^D \rightarrow \mathbb{R}^D$ is the flow map and n denotes the discrete time step. The discrete-time formulation is a natural setting when considering simulation or experimental data. Next, we seek evolutionary equations for a set of observables $\mathbf{g} \in \mathbb{R}^r$, $r < D$, with $g_i : \mathbb{R}^D \rightarrow \mathbb{R}$, $i = 1, \dots, r$. The observables are, in general, functions of the states $\mathbf{x}$. In this study, the observables are selected using an SVD-based compression of the snapshot matrices of the states (achieved by projecting onto the dominant POD modes, as is done with DMD). The central result of the MZ procedure is the GLE (see [15, 16, 18] for detailed derivations), which prescribes the exact and closed set of evolutionary equations for the observables given any initial condition $\mathbf{x}_0$ as:

$$\mathbf{g}_{n+1}(\mathbf{x}_0) = \mathbf{\Omega}_0(\mathbf{g}_n(\mathbf{x}_0)) + \sum_{l=1}^n \mathbf{\Omega}_l(\mathbf{g}_{n-l}(\mathbf{x}_0)) + \mathbf{W}_n(\mathbf{x}_0), \quad (2)$$

where $\mathbf{g}_n : \mathbb{R}^D \rightarrow \mathbb{R}^r$ is the $r \times 1$ vector of functions of the initial state $\mathbf{x}_0$ so that $\mathbf{g}_n(\mathbf{x}_0) \equiv \mathbf{g}(\mathbf{x}(n\Delta t; \mathbf{x}_0)) \equiv \mathbf{g}(\mathbf{F}^n(\mathbf{x}_0))$. The GLE (Eq. 2) states that the vector of observables at time $n+1$ evolves (and is decomposed) according to three parts: (1) a *Markovian* operator: $\mathbf{\Omega}_0 : \mathbb{R}^r \rightarrow \mathbb{R}^r$ which only depends on the observables at the previous time step (n), (2) the *memory kernel*: the series of operators $\mathbf{\Omega}_l : \mathbb{R}^r \rightarrow \mathbb{R}^r$ depending on observables with a time lag l , and (3) the *orthogonal dynamics*: $\mathbf{W}_n : \mathbb{R}^D \rightarrow \mathbb{R}^r$ depending on the full initial state $\mathbf{x}_0$. The above GLE is general for any projection operator (see [17] for a detailed discussion). Using Mori's linear projection [15], which employs the equipped inner product in the L^2 Hilbert space to define the functional projection, results in a linear Markovian form $\mathbf{\Omega}_0(\mathbf{g}_n(\mathbf{x}_0)) = \mathbf{\Omega}_0 \cdot \mathbf{g}_n(\mathbf{x}_0)$, and linear memory kernel $\mathbf{\Omega}_l(\mathbf{g}_{n-l}(\mathbf{x}_0)) = \mathbf{\Omega}_l \cdot \mathbf{g}_{n-l}(\mathbf{x}_0)$, where $\mathbf{\Omega}_l$'s are $r \times r$ matrices. These matrices can be learned by a convex optimization procedure (see Algorithm 1), that incrementally minimizes the residual (orthogonal dynamics) as derived in Lin et al. [18]. In this manuscript, we assume that, in the process of learning the MZ operators with the convex optimization scheme derived in [18], $\mathbf{W}_n$ is a small and negligible residual term (i.e. the model represents the observables evolving in the projected space).

III. Mori-Zwanzig Modal Decomposition

Following our recent work [22], we extend the data-driven MZ approach by constructing a modal analysis technique which includes MZ memory kernels. Similar to DMD, we use SVD compression to provide a tractable method for extracting the modes and spectrum in the state space from a reduced set of observables defined using POD modes. Rather than starting with the standard DMD assumption that the evolution of state variables is linear (i.e., $\mathbf{x}_{n+1} = \mathbf{A}\mathbf{x}_n$), with MZMD, we start from the discrete-time GLE (2) and assume that the higher-order orthogonal dynamics are negligible (i.e. we are evolving in the projected space). This is formalized with the Mori's projector and D state observables selected as $\pi_i \triangleq \pi_i(\mathbf{x}) = x_i$. In this case, the memory kernels are minimizing the residual associated with the linear operator ($\mathbf{\Omega}_0^{(x)}$) of DMD, thus they are attempting to account for the what DMD is missing (i.e. transient and intermittent dynamics). Assuming that only k memory terms are sufficient so that the higher-order orthogonal dynamics are negligible, Eq. (2) becomes

$$\mathbf{x}_{n+1} = \mathbf{\Omega}_0^{(x)} \mathbf{x}_n + \dots + \mathbf{\Omega}_k^{(x)} \mathbf{x}_{n-k}, \quad (3)$$

where the linear MZ operators $\mathbf{\Omega}_i^{(x)}$ are acting in state space.

Now, decomposing the snapshot matrix using an SVD, $\mathbf{X} = [\mathbf{x}_0, \dots, \mathbf{x}_\tau] \approx \mathbf{U}_r \mathbf{\Sigma}_r \mathbf{V}_r^*$, the reduced observables are selected as $\mathbf{g}(\mathbf{x}_n) \triangleq \mathbf{U}_r^* \mathbf{x}_n$. Therefore, we can write Eq. 3 as

$$\mathbf{g}_{n+1} = \mathbf{\Omega}_0^{(g)} \mathbf{g}_n + \dots + \mathbf{\Omega}_k^{(g)} \mathbf{g}_{n-k}, \quad (4)$$

where $\Omega_i^{(g)} = U_r^* \Omega_i^{(x)} U_r$ is the memory kernel $\Omega_i^{(x)}$ projected onto the POD modes. We establish the relationship between the modes of 3 and 4 in two parts; first a full SVD ($r = D$) is used to establish the equivalence, then a truncated SVD is used ($r < D$) resulting in approximations of the modes of the full state GLE (eq. 3). We note that by adding memory kernels there is now a multiplicative interaction between the Ω_i 's, as seen in the long time dynamics described below. Furthermore, the modal analysis can be obtained through the associated block companion matrices

$$C_x = \begin{bmatrix} \Omega_0^{(x)} & \Omega_1^{(x)} & \dots & \Omega_k^{(x)} \\ I & \mathbf{0} & \dots & \mathbf{0} \\ & \ddots & & \\ \mathbf{0} & \dots & I & \mathbf{0} \end{bmatrix} \quad \text{and} \quad C_g = \begin{bmatrix} \Omega_0^{(g)} & \Omega_1^{(g)} & \dots & \Omega_k^{(g)} \\ I & \mathbf{0} & \dots & \mathbf{0} \\ & \ddots & & \\ \mathbf{0} & \dots & I & \mathbf{0} \end{bmatrix}, \quad (5)$$

where I and $\mathbf{0}$ are the appropriately sized identity and zero matrices respectively. This shares a similar structure to Higher-Order DMD [8], however, in this work the higher order terms $\Omega_{k>0}^{(g)}$ are part of the MZ memory kernel, which satisfies the generalized fluctuation-dissipation relationship; each learned using the algorithms described in [18] and in Algorithm 1.

The solutions to Eq. 3 are therefore given by

$$\mathbf{x}_n = P_0 C_x^n \mathbf{z}_0,$$

where $\mathbf{z}_0 = [\mathbf{x}_0, \dots, \mathbf{x}_{-k}]^T$ and $P_0 = [I \ \mathbf{0} \ \dots \ \mathbf{0}]$, and the solution for Eq. 4, is $\mathbf{g}_n = P_0 C_g^n \tilde{\mathbf{z}}_0$ where $\tilde{\mathbf{z}}_0 = [\mathbf{g}_0, \dots, \mathbf{g}_{-k}]^T$. Thus, the solutions are fully characterized by the eigenpairs of the associated companion matrices (C_x , and C_g), such as described below in equations 6 and 7.

$$\mathbf{g}_n = P_0 \sum_i a_i \lambda_i^n \mathbf{w}_i = \sum_i a_i \lambda_i^n \mathbf{w}_i^0, \quad \mathbf{w}^0 \equiv P_0 \mathbf{w}. \quad (6)$$

The eigenvectors identify the large-scale coherent structures and their associated eigenvalues determine the temporal evolution as seen in the basis expansion about the eigenvectors (modes) ϕ_i , as seen below in Eq. 7;

$$\mathbf{x}_n = P_0 \sum_i a_i \lambda_i^n \phi_i = \sum_i a_i \lambda_i^n \phi_i^0, \quad \phi^0 \equiv P_0 \phi. \quad (7)$$

However, working with the matrix C_x can be computationally infeasible for large systems. Thus, we will instead use C_g for computing the eigendecomposition (and MZ memory terms). However, in order to make statements about the state space MZ model (Eq. 3) we need to relate the matrices C_x , and C_g . In what follows, we show a similarity transformation between the two provides a way to relate the modes and eigenvalues.

We show the relationship between the eigenpair of the full system Eq. 3 with that of C_g . A similar result is shown for DMD [3, 27], which is true in what follows when the number of memory terms is set to zero. Let $(\lambda, \mathbf{w})$ be an eigenpair of C_g with ($r = D$). Then (λ, ϕ) is an eigenpair of C_x where $\phi = [\phi^0, (1/\lambda)\phi^0, \dots, (1/\lambda^k)\phi^0]^T$, and $\phi^0 = U\mathbf{w}^0$. Starting with $X = U\Sigma V^*$ be the full SVD of the snapshot matrix ($r = D$). Then, expanding $C_g \mathbf{w} = \lambda \mathbf{w}$, the associated nonlinear eigenvalue problem ([28]) of C_g satisfies

$$\Omega_k^{(g)} \mathbf{w}^0 + \lambda \Omega_{k-1}^{(g)} \mathbf{w}^0 + \dots + \lambda^k \Omega_0^{(g)} \mathbf{w}^0 = \lambda^{k+1} \mathbf{w}^0.$$

Substituting $\Omega_i^{(g)} = U^* \Omega_i^{(x)} U$ then left multiplication by U results in

$$\Omega_k^{(x)} (U\mathbf{w}^0) + \lambda \Omega_{k-1}^{(x)} (U\mathbf{w}^0) + \dots + \lambda^k \Omega_0^{(x)} (U\mathbf{w}^0) = \lambda^{k+1} (U\mathbf{w}^0).$$

This is the associated nonlinear eigenvalue problem of C_x , thus (λ, ϕ) is an eigenpair of C_x . The above formulation of MZMD extends the data-driven MZ method and establishes the relationship between the modes and spectrum of 3 and eigenpairs of 4.

In practice, a truncated SVD is computed ($r < D$) [29] to obtain a suitable low rank approximation. In this case, $U_r U_r^*$ forms an orthogonal projection operator, projecting onto the first r POD modes of the snapshot matrix. This fact together with the above argument allows us to approximate the modes and spectrum of the state space GLE 3 from the eigenpair of the reduced observable GLE 4, which is otherwise intractable for large systems. Figure 1 illustrates the

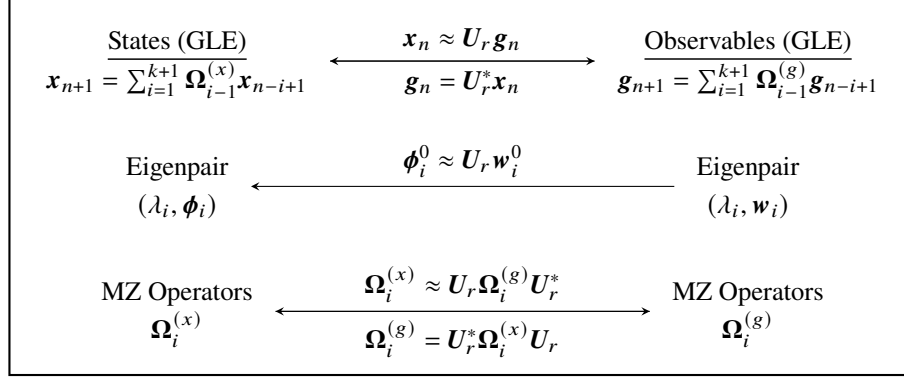


Figure 1 MZMD summary

relationships between observables, eigenpairs, and operators in these two spaces. The MZMD modes are consequently $\boldsymbol{\Phi} \equiv [\boldsymbol{\phi}_1^0, \dots, \boldsymbol{\phi}_{rk}^0]$. With the standard DMD approach the amplitudes are then found as $\mathbf{a} = \boldsymbol{\Phi}^{-1} \mathbf{z}_0$, which can be done here as well with $\mathbf{a} = \tilde{\boldsymbol{\Phi}}^\dagger \mathbf{z}_0 \approx (\mathbf{W})^{-1} \tilde{\mathbf{z}}_0$, where $\tilde{\boldsymbol{\Phi}} \equiv [\boldsymbol{\phi}_1, \dots, \boldsymbol{\phi}_{rk}]$ and $\mathbf{W}$ the is eigenvector matrix of $\mathbf{C}_g$ (i.e. $\mathbf{z}_0$ expressed in basis of eigenvectors). This provides a scalable modal analysis technique that improves upon DMD by adding hysteresis effects with the Mori-Zwanzig formalism. It is useful to point out that when there is no memory present in MZMD, it is equivalent to DMD. Indeed, it has been established that MZ without memory terms is equivalent to EDMD [18]. Thus, MZMD generalizes DMD using the MZ formalism to include memory effects within the modes and spectrum.

In this work we compute the amplitudes with a least squares approximation as is done in other works (see Le Clainche and Vega [8] and references therein) and described below in Eq. 8. This approach utilizes the full range of training data, as opposed to computing $\mathbf{a}$ by only expressing the initial conditions the in basis of eigenvectors, as described above. Following equation 6, we see the following relations of the amplitudes, modes and training data.

$$\begin{cases} \mathbf{g}_0 = \boldsymbol{\Phi}^0 \mathbf{a} \\ \mathbf{g}_1 = \boldsymbol{\Phi}^0 \mathbf{\Lambda} \mathbf{a} \\ \vdots \\ \mathbf{g}_n = \boldsymbol{\Phi}^0 \mathbf{\Lambda}^n \mathbf{a} \end{cases} \iff \begin{bmatrix} \boldsymbol{\Phi}^0 \\ \boldsymbol{\Phi}^0 \mathbf{\Lambda} \\ \vdots \\ \boldsymbol{\Phi}^0 \mathbf{\Lambda}^n \end{bmatrix} \mathbf{a} = \begin{bmatrix} \mathbf{g}_0 \\ \mathbf{g}_1 \\ \vdots \\ \mathbf{g}_n \end{bmatrix} \quad (8)$$

Solving the over-determined system of equations 8 with least squares can provide a best fit amplitude vector for the entire training set. This can improve the ability to reconstruct and predict the state space measurements $\mathbf{x}(t)$.

IV. 2D Transient Flow over Cylinder

In this section we demonstrate the notable ability of MZMD to reconstruct the state space of a 2D transient flow over cylinder by adding MZ memory. This is a flow regime in which DMD is known to fail, however, we also demonstrate this in the following analysis. We generate transient flow data using the open source immersed boundary method of Colonius and Taira [25] with a Reynolds number of $Re = 200$ and $dt = 0.01$ (so that the CFL condition is satisfied). This flow evolves from an unstable equilibrium point to a heteroclinic orbit (see figure 2). We compare DMD and MZMD on their ability to reconstruct the evolution of the vorticity states $\mathbf{x}$ of this flow. We show that adding MZ memory introduces MZMD modes that capture the transient dynamics as well as stabilizing the modes with eigenvalues near the unit circle as compared to DMD.

We perform a parameter sweep over both r (the selected spatial complexity, i.e. the SVD truncation) and k (the number of memory terms in MZMD). We measure the relative reconstruction errors with the mean squared error (MSE). This error analysis is reported in Fig. 3. We see that there is a significant improvement by adding MZ memory. As reported in previous work [22], this is achieved at nearly the same computational cost as DMD. We perform this same analysis for short time future state prediction (relative generalization errors), this is reported in Fig. 4. From this error analysis, we see that MZMD significantly improves upon DMD with respect to both reconstruction and generalization errors. In Fig. 5 we show where in the 2D domain MZMD and DMD has point-wise errors (averaged over time). We

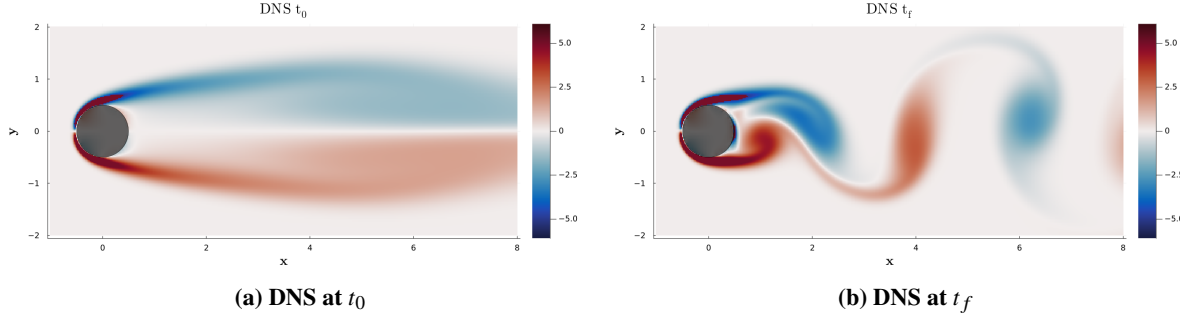


Figure 2 (a) Snapshot of the initial unstable equilibrium point. (b) Snapshot of the stable heteroclinic orbit representing von Kármán vortex street

next investigate why MZMD has such an improved performance for this transient flow. We see in Fig. 15 that the relative contribution of the memory terms is decaying, and nontrivial up to $k \approx 20$ (which can be used as a selection criterion).

Next, we perform modal analysis, looking at both the eigenvalues and amplitudes of modes. In Fig. 6 we report the eigenvalues from DMD vs. MZMD. We see that DMD contains unstable modes that will grow exponentially over time in an attempt to capture global linear modes that evolve from the unstable equilibrium to the heteroclinic orbit. This is seen in the eigenvalues that lie outside the unit circle with magnitude greater than unity. However, with MZMD, although not all the modes are perfectly periodically stable, the eigenvalues associated with periodic modes have been pushed closer to the unit circle. In Fig. 7 we show the amplitudes of the modes vs. frequencies. This shows that MZMD introduces low amplitude transient modes as well as larger amplitudes high frequency modes in order to capture the transient flow. Furthermore, we see the introduction of many transient modes with MZMD, with short time dynamic evolution, so that in combination they reproduce the flow field (as seen in fig. 8). In Fig. 9 we report the dominant modes, as arranged in descending order from their largest to smallest amplitudes, however further investigation into these modes are required in order to understand which are most responsible for the transition. Therefore, we see that MZMD improves stability as well as introduces modes that capture short time transient behavior, whereas DMD is not able to reconstruct the flow at all (as seen in the point-wise error figure 5).

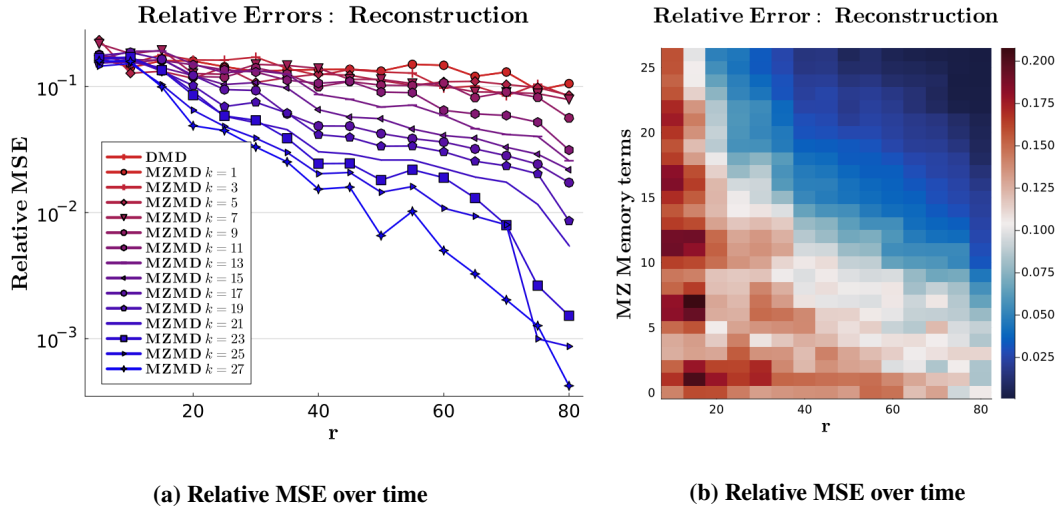


Figure 3 (a) MSE of reconstruction errors, in log scale showing improvement with adding MZ memory in MZMD. (b) MSE of reconstruction errors contours.

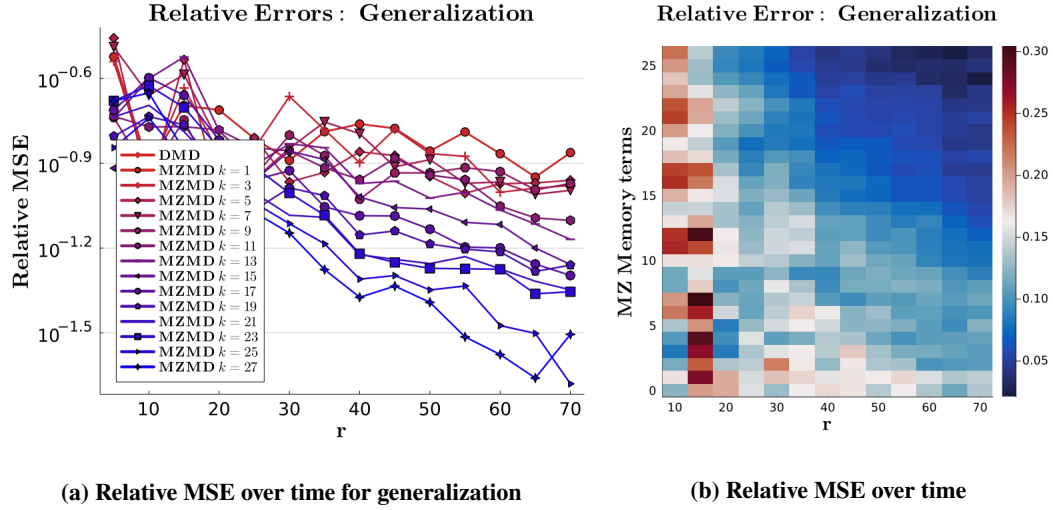


Figure 4 (a) MSE of short time future state prediction errors, in log scale showing improvement with adding MZ memory in MZMD. (b) MSE of short time future state prediction errors contours.

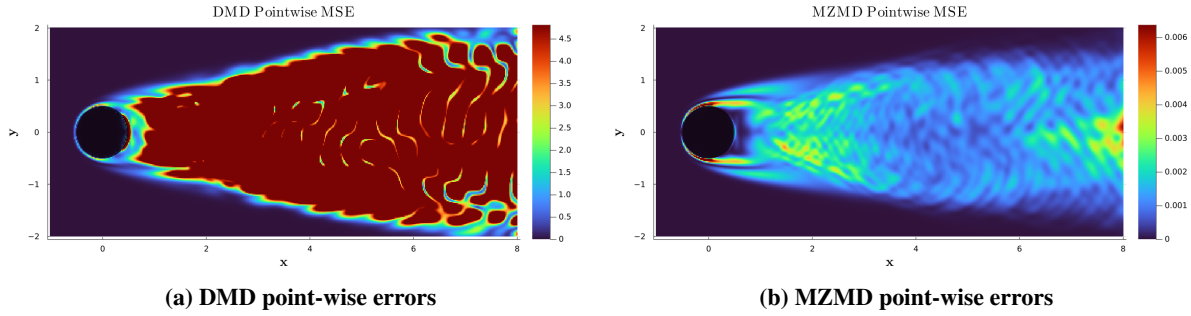


Figure 5 (a) point-wise MSE reconstruction errors (averaged over time of the training set) of DMD, showing large errors which accumulate due to exponential growth of modes. (b) point-wise MSE reconstruction errors (averaged over time of the training set) of MZMD, showing much smaller errors in the downstream region of the cylinder. This is a several orders of magnitude improvement over DMD on reconstructing the transient flow

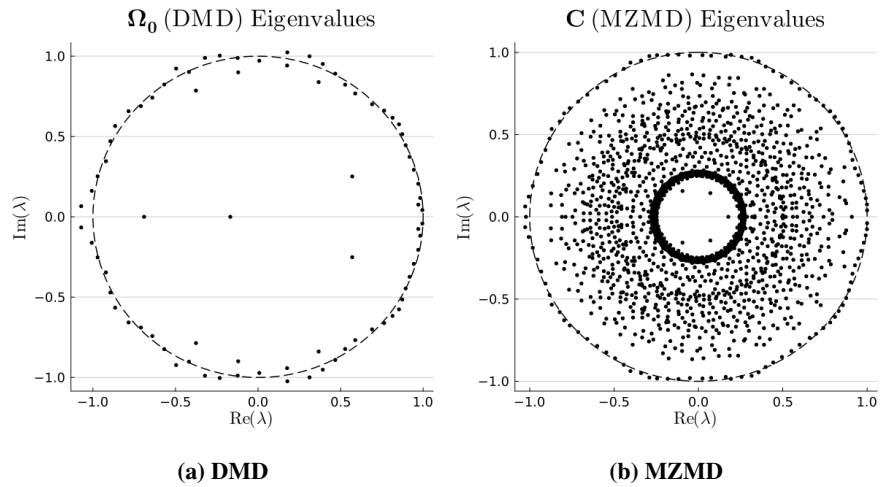


Figure 6 (a) Eigenvalues of DMD, showing unstable modes. (b) Eigenvalues of MZMD describing temporal evolution of modes, containing transient dynamics, and improves stability of unstable modes

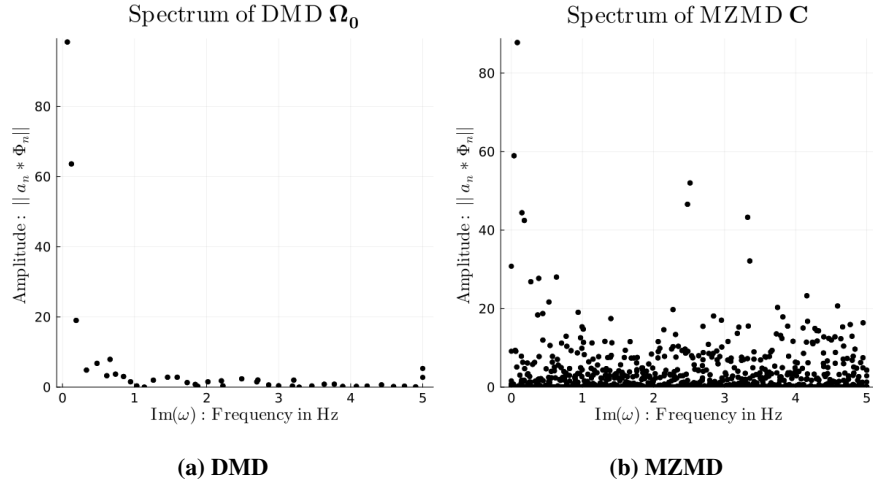


Figure 7 (a) Amplitudes of DMD vs. frequencies (b) Amplitudes of MZMD vs. frequencies

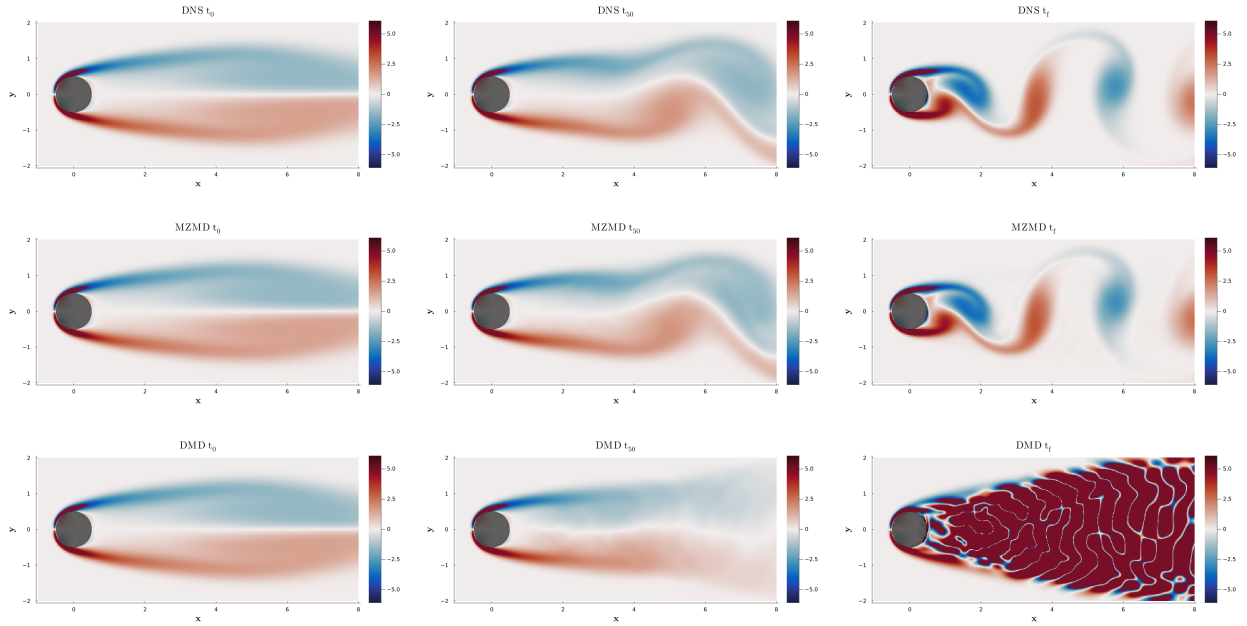


Figure 8 Showing snapshots of the evolution of DNS (top), MZMD (middle), and DMD (bottom) predictions of vorticity states, demonstrating the ability of MZMD to capture the transient behavior, $r = 80$, $k = 27$.

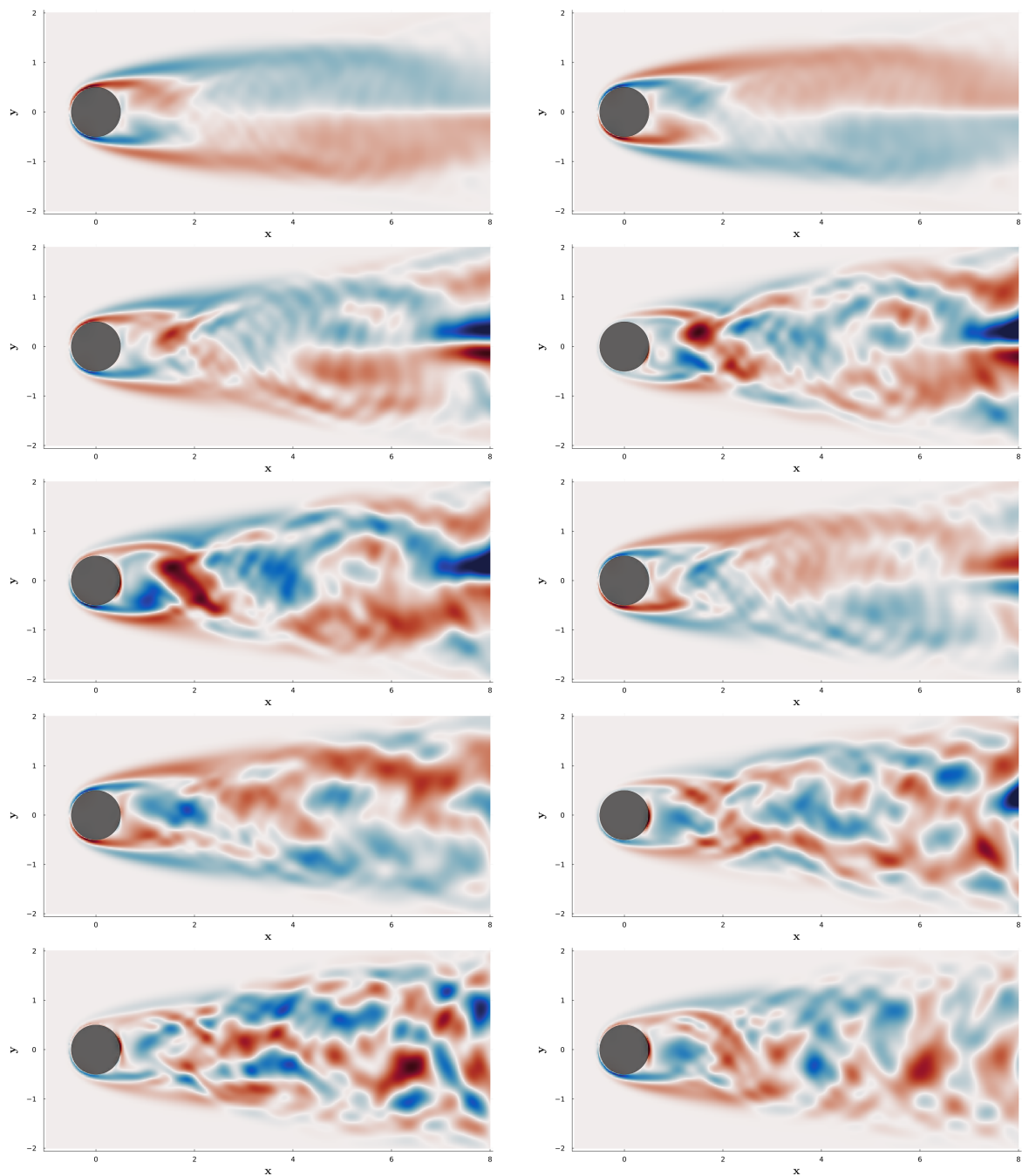


Figure 9 Dominant MZMD modes, arranged in their amplitude content from the top down.

V. Hypersonic Boundary Layer Transition

High-speed laminar–turbulent boundary-layer transition is highly complex and remains a major unresolved topic in fluid dynamics. Understanding high-speed boundary-layer transition is necessary in order to develop reliable transition prediction and flow control methods. This is important for the next-generation design and safe operation of advanced high-speed vehicles. Transition to turbulence leads to significant increases in skin friction (drag) and heat transfer. In the transitional regime nonlinear interactions can also lead to locally very high skin-friction and heat transfer values ("hot" streaks) far exceeding the laminar values. Wind tunnel experiments [30] as well as DNS [31] showed the development of these so-called "hot" streaks in the nonlinear transition regime. Therefore, understanding the dominant physical mechanisms leading to laminar-turbulent boundary-layer transition is vital for the design and safe operation of a high-speed vehicle. The transition process is inherently nonlinear, yet the current state-of-the-art transition prediction models used for the design of high-speed vehicles are based on linear theory only. Therefore, reduced-order modeling can significantly aid in the understanding of the dominant nonlinear transition mechanisms and has the potential to improve future transition prediction methodologies. Furthermore, the transition process presents a significant challenge for modal analysis techniques, as they contain strong nonlinearities localized within regions of the flow. In the next section we demonstrate that using MZ memory in MZMD significantly improves the ability to reconstruct and predict these strong nonlinearities present for a transitional $M = 5.3$ boundary-layer over an ogive. We show that MZMD achieves this by adding low-amplitude transient modes that attempt to capture the intermittent dynamics present in the transition region of the flow.

VI. Results: Tangent Ogive at Mach 5.3

In this section we give the same analysis discussed in the previous sections but now applied to a transitional $M = 5.3$ boundary-layer of a tangent ogive with a blunted nose tip and at zero angle of attack. This flow is a representative example of a high-dimensional complex dynamical system that exhibits strong nonlinearities in the region of transition to turbulence. In this region, intermittent dynamics are present, a characteristic of turbulent flows. The transition simulations are carried out at the University of Arizona [26, 31, 32]. Experiments for the tangent ogive geometry were carried out at the hypersonic wind tunnel (H2K) at the German Aerospace Center (DLR) in Cologne. We use the both the pressure disturbances at the wall for the state space measurements $\mathbf{x}(t)$ in MZMD. The geometry and a snapshot the pressure disturbances of this flow are shown in Fig 10a.

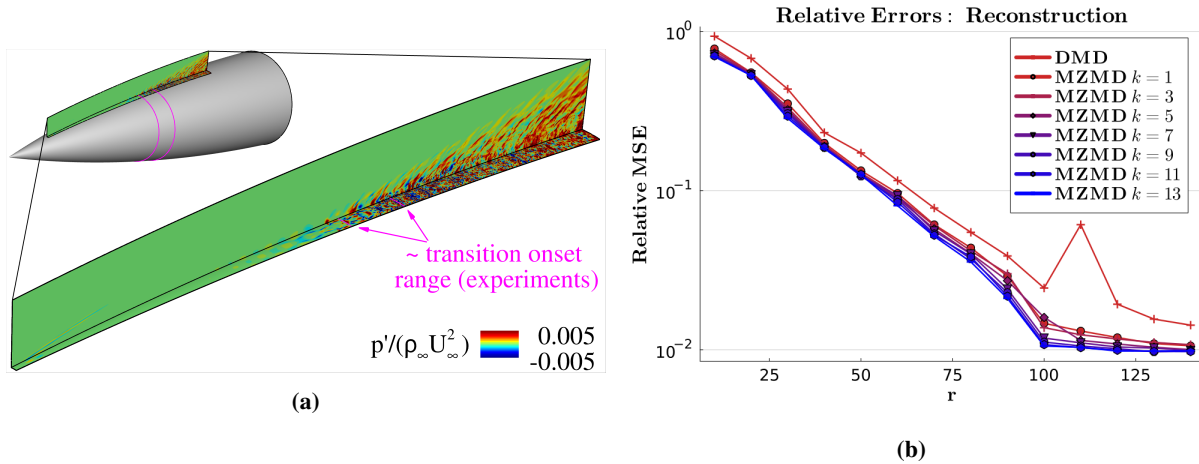


Figure 10 Instantaneous pressure disturbance contours at the wall and in a wall-normal plane for a tangent ogive at Mach 5 (a). Relative errors of MZMD (b) applied to ogive.

To summarize the results of this section, MZMD improves the ability to reconstruct the pressure disturbance at the wall (see Fig. 10). This improvement is especially apparent in the point-wise MSE (as seen in Fig. 13), averaged over time, as seen in the transitional region of the flow. MZMD achieves this by introduction of small, yet non-trivial, amplitude transient modes as seen in Figs. 11, and 12. These transients modes are accounting for intermittent dynamics, a key characteristic of transitional and turbulent boundary layers. Therefore, the MZMD modes represent a more accurate descriptions of the dominant transition mechanism (as seen in Fig. 16) present in this flow, which contains

strong nonlinearities within the transitional regime. Furthermore, this is achieved at nearly the same computational cost as DMD [22], and MZMD is easily portable to existing codes (as described in Appendix A). In figure 16 and 12, we see the dominant modes are cascading from large scale structures to higher harmonic modes (with shorter wavelengths), similar with the cascade to turbulence. For this flow regime, these higher harmonics are also being lifted further downstream with greater frequencies and shorter wavelengths, as is common for transitional flows.

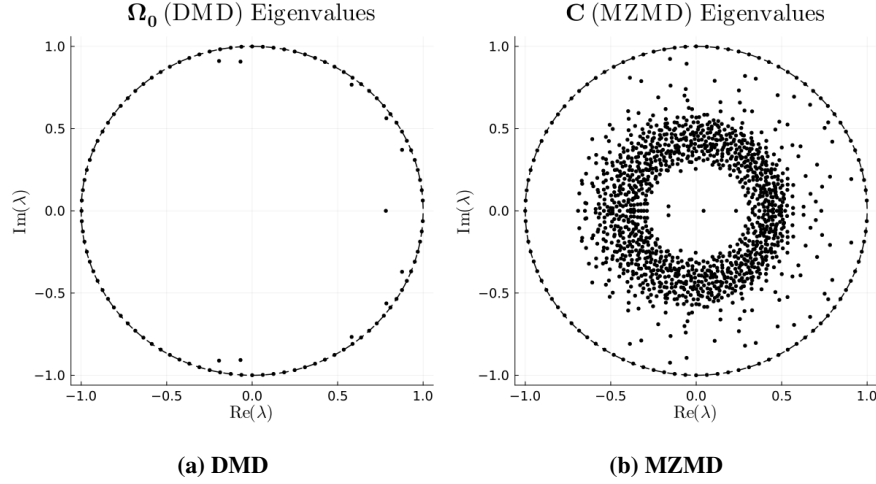


Figure 11 Eigenvalues of DMD (a) and MZMD (b). These describe the temporal evolution of modes. The presence of transient dynamics is apparent from eigenvalues contained within the unit circle.

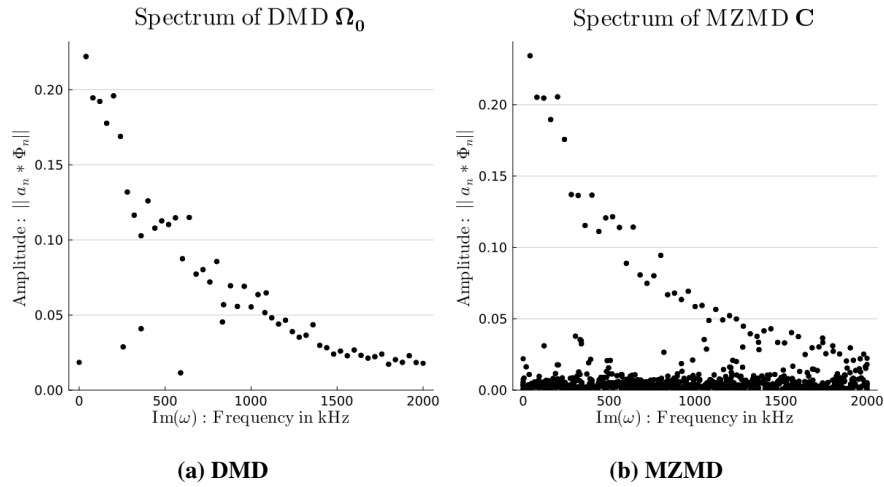


Figure 12 Amplitudes vs. frequencies of DMD (a) and MZMD (b). MZMD introduces small, but nontrivial transient modes to improve reconstruction within nonlinear regions. Both DMD and MZMD capture the cascade from dominant large scale modes with low frequency to small scale high frequency modes, a key characteristic of turbulent flows.

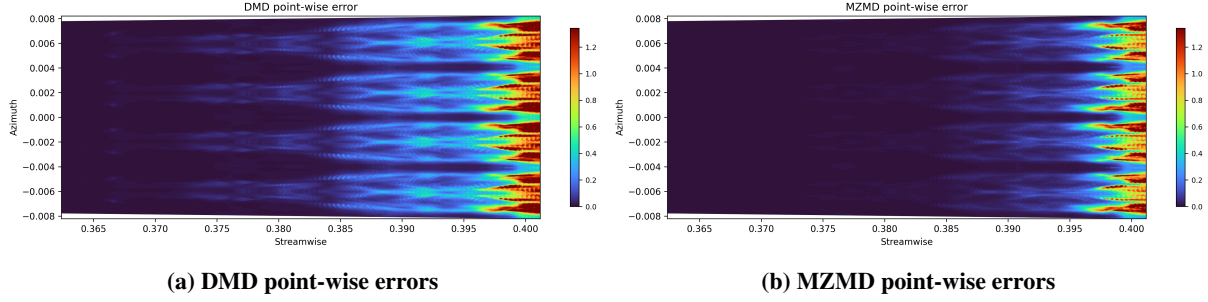


Figure 13 The point-wise MSE, averaged over time, zooming in on the transition region to show the errors present in the future state predictions of DMD (a) and MZMD (b). We see that DMD has larger errors within the transition region as compared to DMD. Thus MZMD improves future state prediction in regions with strong nonlinearities. These figures are zoomed in near the transition region, as highlighted in the transition onset region in figure 10).

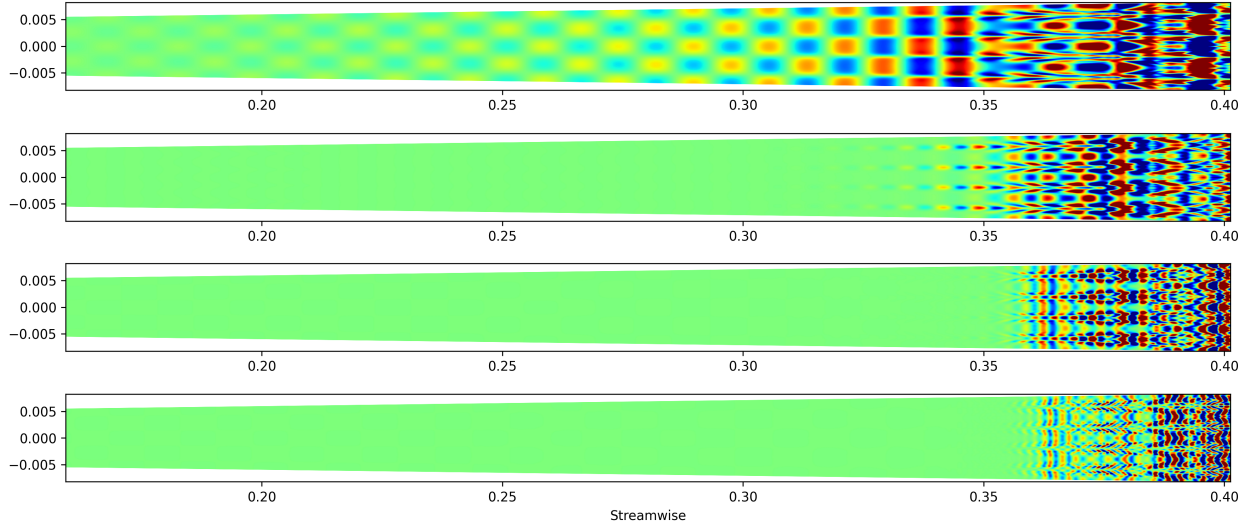


Figure 14 Dominant MZMD modes (unrolled). The dominant modes (top) contains the large scale structure of the exponential amplification of oblique waves that eventually saturates downstream and captures the onset of transition. The higher harmonic mode (second row) cascading further downstream locations with higher frequencies and smaller wavelengths (as seen in last row).

VII. Conclusion

In this manuscript, we approached two exemplary flows that represent significant challenges for modal analysis techniques; namely, flows with strong localized nonlinearities and transient flows. We showed that adding MZ memory in MZMD represents an efficient generalization of DMD that significantly improves its ability to reconstruct and predict both transient flows and strong nonlinearities by including transient (or intermittent) modes. The first flow considered was a 2D transient flow over a Cylinder. It is well known that DMD fails to reconstruct this flow. We showed that MZMD is able to reconstruct this flow and performs order of magnitude better than DMD with respect to relative MSE. MZMD achieves this by introducing MZ memory, which accounts for hysteresis effects which play a large role. We showed that by including MZ memory, this introduces transient modes within the modal decomposition. This allows it to captures the short time transient dynamics. The next flow we considered was the hypersonic flow over an ogive that exhibits laminar-turbulent transition. This flows contains strong nonlinearities in the transition region which DMD struggles to capture. We provided numerical evidence that demonstrates MZMD, by introducing intermittent modes within the transition region, improves the ability to predict these nonlinear mechanisms, which contain localized intermittent dynamics within the transition region. Therefore MZMD represents more accurate modal decomposition

and spectral analysis technique.

Many future research directions can be explored by adding MZ memory. The relationship established in this work makes it straightforward to implement MZ memory kernels into existing DMD codes. For example, DMD with control and online DMD [33] could be improved using MZMD. Furthermore, adding physical structure within the MZ operators is also possible using regression as a projection operator [17], to enforce physical constraints and conservation laws [34, 35].

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Declaration of interests. The authors report no conflict of interest.

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A. Algorithms

In this appendix, we provide an overview of the MZMD algorithm (1) and establish connections with the DMD algorithm (2) to establish consistent referencing and notation alignment. In MZMD 1 we first obtain the snapshot data as is done in POD and DMD but we also need to include some past history for which the parameter k is used. In both algorithms, we can interpret the SVD as a linear auto-encoder as a automatic technique for selecting observables from data (as a lossy data compression to avoid the intractable computations of the full two time covariance matrix $\tilde{C}_k = X_k \cdot X_1^T$). This is equivalent to performing a low rank approximation of the full $\tilde{C}_k$ by projecting onto the POD modes (which is what is done in the standard DMD algorithm). This equivalence can be seen by replacing G_k with $U_r^* X_k$ when computing C_k , to see that $C_k = U_r^* X_k (U_r^* X_0)^T = U_r^* X_k X_0^T U_r \approx \tilde{C}_k$. Hence C_k is a low rank approximation of the full $\tilde{C}_k$ by projecting onto the first r dominant POD modes U_r .

Algorithm 1 MZMD

- 1: Select the number of memory terms k
 - 2: Given $T + 1$ snapshots of data: $X = [x_0, \dots, x_T]$
 - 3: Compute truncated SVD $X \approx U_r \Sigma_r V_r^*$
 - 4: Project onto POD modes $G = U_r^* X$
 - 5: Collect snapshots over k time delays: $G_0 = [g_0, g_1, \dots, g_{T-k}], G_1 = [g_1, g_2, \dots, g_{T-k+1}], \dots, G_k = [g_k, g_{k+1}, \dots, g_T]$
 - 6: $C_0 = G_0 G_0^T$
 - 7: **for** $i \leftarrow 1, \dots, k$ **do**
 - 8: $C_i = G_i G_0^T$
 - 9: **end for**
 - 10: $\Omega_0 = C_1 C_0^{-1}$
 - 11: **for** $i \leftarrow 1, \dots, k$ **do**
 - 12: $\Omega_i = [C_{i+1} - \sum_{l=0}^{i-1} \Omega_l C_{i-l}] C_0^{-1}$
 - 13: **end for**
 - 14: - Form companion matrix C_g , then compute eigendecomposition $C_g W = W \Lambda$
 - 15: - **MZMD modes**: $\phi_i^0 = U_r w_i^0$, where $w_i^0 = P_0 w_i$ and $P_0 = [I \ 0 \ \dots \ 0]$.
 - 16: - Compute amplitudes: $a_i = \Phi^\dagger [x_0, x_1, \dots, x_k]^T$
 - 17: - $x_{n+1} \approx \sum_i^{r,k} a_i \lambda_i^{n+1} \phi_i^0$ (selected modes prediction)
-

Algorithm 2 DMD

- 1: Given $T + 1$ snapshots of data: $X = [x_0, \dots, x_T]$
 - 2: $X \approx U_r \Sigma_r V_r^*$ Truncated SVD
 - 3: Low rank projection $G = U_r^* X$
 - 4: Collect snapshots over k time delays: $G_0 = [g_0, g_1, \dots, g_{T-1}], G_1 = [g_1, g_2, \dots, g_T]$
 - 5: $C_0 = G_0 G_0^T$
 - 6: $C_1 = G_1 G_0^T$
 - 7: $K = C_1 C_0^\dagger$
 - 8: - Compute eigendecomposition $KW = W\Lambda$
 - 9: - **DMD modes**: $\phi_i = U_r w_i$
 - 10: - Compute amplitudes: $a_i = \Phi^\dagger x_0$
 - 11: - $x_{n+1} \approx \sum_i^r a_i \lambda_i^{n+1} \phi_i$
-

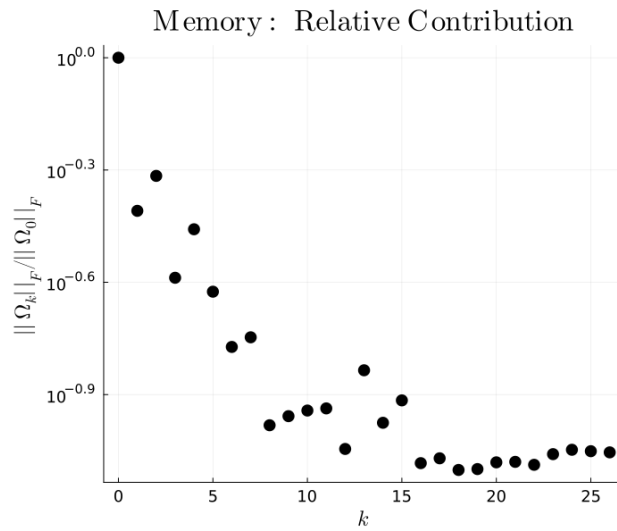


Figure 15 Relative contribution of the MZ memory kernels with respect to the Markovian (DMD) operator.

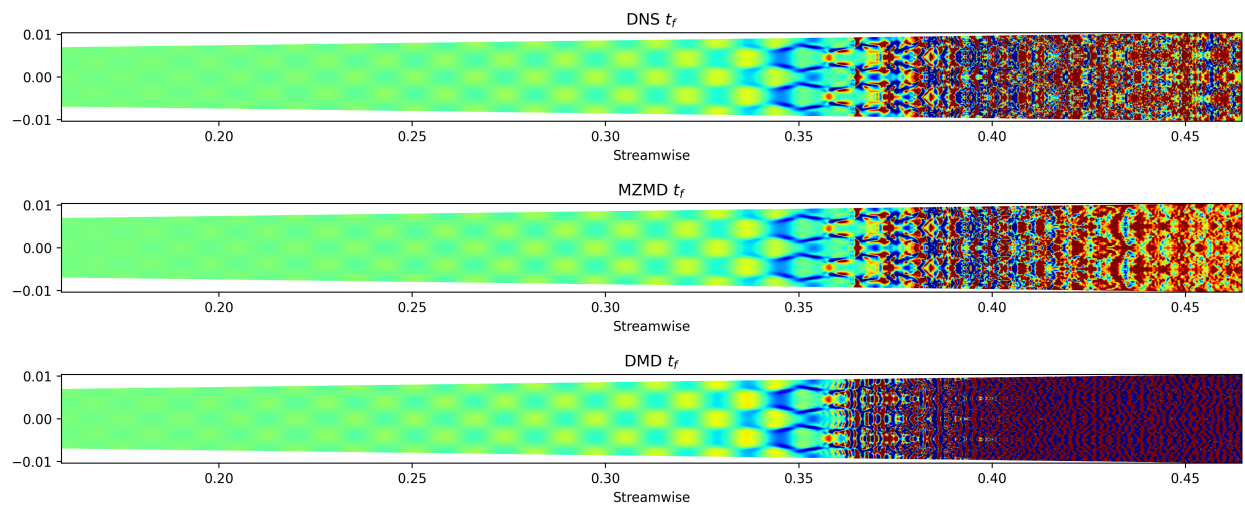


Figure 16 Unrolled snapshots of future state prediction after 50 time steps, showing a qualitative description of the error in DMD vs. MZMD ($r = 200$, $k = 7$).